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Equity-Centered Development of a Community Health Worker Intervention to Improve Engagement in Care for ADHD

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Abstract

Objective: Barriers to engagement perpetuate inequities in ADHD care for children. Community Health Worker (CHW) interventions could improve treatment engagement and reduce inequities. We describe the development of a novel, manualized, 5-module CHW intervention for ADHD called START (SupporT for ADHD and Related Treatment) to increase engagement with ADHD care. **Method:** We developed START based on perspectives from racial and ethnic minoritized families with feedback from a community advisory board, and conducted a single arm, adaptive pilot feasibility study of START for primary caregivers of children aged 6 to 12 years with newly diagnosed ADHD. We collected process data, pre- and post-intervention survey data, and qualitative data via caregiver exit interviews. **Results:** Eighteen families enrolled, 15 completed START, and 16 completed study questionnaires. Median intervention duration was six sessions (IQR = 5–6) over 5 weeks (IQR = 4–8). All caregivers reported satisfaction with the quality of the intervention and materials provided and reported they would recommend START to another caregiver. Most caregivers reported that START made them more likely to consider medications (69%), therapy (75%), and school services (88%) for their ADHD. From pre- to post-intervention, median number of outpatient treatments increased from 1.0 (IQR = 1.5) to 2.0 (IQR = 0.5; $p = .0034$), ADHD knowledge increased ($p < .0001$), and median scores for 16 of 17 intervention objectives increased ($p < .05$). **Conclusion:** START showed excellent feasibility and acceptability and treatment utilization increased post-intervention. We refined START based on feedback from caregivers, including enhanced personalization and expanding to six modules, to prepare for testing in a clinical trial. (*J. of Att. Dis. 2026; 30(5) 577-589*)

Keywords
community health workers, ADHD, child, health inequities, minority health

Introduction

ADHD is a neurobiological condition affecting about 6% to 8% of children and adolescents (Polanczyk et al., 2007; Visser et al., 2014) characterized by inattention and/or hyperactivity and impulsivity (American Psychiatric Association, 2013). ADHD can have lifelong consequences, increasing the risk for educational and occupational underachievement, family and peer conflict, justice involvement, secondary psychopathology, substance use disorders, accidental injury, and premature death including by suicide (Biederman et al., 2012; Impey & Heun, 2012; Sibley et al., 2011). Treatment for ADHD, which includes a combination of FDA-approved medication, behavioral therapy, and school accommodations, is effective at reducing symptoms and can improve ADHD outcomes (Arnold et al., 2020; Chang et al., 2020; Shaw et al., 2012; Shoval et al., 2021).

However, there are well-documented racial, ethnic, and socioeconomic inequities in access to and engagement in

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evidence-based ADHD treatment (Coker et al., 2016; Shi et al., 2021; Yang et al., 2022). Compared to White youth, Black, Hispanic, and Asian children with ADHD receive less treatment, including fewer visits with a healthcare provider and lower medication utilization (Coker et al., 2016; Shi et al., 2021; Yang et al., 2022). Barriers to care include logistical difficulties obtaining and continuing care (e.g., finances, insurance, transportation, and lack of child care) as well as "silent" barriers such as stigma, discrimination, and implicit bias that affect families' experiences with engaging in care (Cummings et al., 2017; Miller et al., 2009; Shi et al., 2021).

Community Health Worker (CHW) interventions have been successfully deployed for other pediatric medical conditions to improve treatment engagement and outcomes while reducing inequities (Coker et al., 2023; Fox et al., 2007; Harries et al., 2023; Volerman & Press, 2022). CHWs are interventionists without formal health training who have shared experiences with the people they serve, such as cultural congruence. They can improve the availability of culturally and linguistically appropriate care, reduce costs and stigma associated with treatment, and address social and structural barriers to care (Barnett et al., 2018). For example, CHW interventions for asthma can improve controller use, reduce emergency department visits and hospitalizations, and improve clinical outcomes (Fox et al., 2007). Despite their promise, CHW interventions have not been well studied for ADHD treatment. A few studies showed evidence that Bachelor's-level care managers could help support engagement in ADHD care, (Fiks et al., 2017; Guevara et al., 2021; Silverstein et al., 2015; Waxmonsky et al., 2023) and one study reported successful implementation of paraprofessional-delivered behavioral parent training for ADHD with seven parents (Chacko et al., 2020). These studies provide initial support for the idea that CHWs could play an important role in ADHD care for underserved families.

Thus, we sought to develop a CHW intervention to reduce inequities in receipt of ADHD care. We used the Six Stages of Engagement in ADHD Care conceptual framework to guide intervention development (Spencer et al., 2021). The Six Stages framework is a family- and equity-centered framework of engagement in ADHD care based on themes from in-depth interviews with families experiencing health inequities, including low-income, racial and ethnic minoritized, and non-English-Speaking families (Spencer et al., 2021). The model describes the process of engaging in care from the perspective of families experiencing inequities in care, and provided a framework for the equity- and family-centered development of our novel intervention called START (SupporT for ADHD and Related Treatment). Here, we describe the development, pilot testing, and refinement of START to prepare for testing in a clinical trial.

Methods

Our process of intervention development included a) consultation with a community advisory board (CAB); b) intervention target selection based on the Six Stages framework; c) intervention content development based on our previous qualitative work and feedback from the CAB; d) Community Health Worker (CHW) training; e) intervention feasibility testing; and f) intervention refinement.

Community Advisory Board

We assembled a Community Advisory Board (CAB) consisting of racial and ethnic minoritized caregivers of children with ADHD, public school staff, pediatric clinical providers and leaders, child mental health clinicians, and child mental health equity researchers to help guide the intervention development. The CAB met twice, in December 2018 (16 attendees) and July 2019 (22 attendees). CAB meetings were not audio recorded, but meeting minutes were taken by research staff. The first CAB meeting focused on discussing preliminary themes from the qualitative work that informed intervention development. The second CAB meeting sought feedback on proposed content and delivery structure of the intervention. CAB meeting agendas are included as supplements (Supplemental 1a and 1b).

Intervention Target Selection

We selected intervention targets based on the Six Stages of Engagement, a family-centered framework centering the experiences of families facing health inequities that has been previously described (Spencer et al., 2021). Selected intervention targets by stage (Figure 1) included: (1) promoting knowledge and acceptance of the ADHD diagnosis; (2) acknowledging and confronting stigma about ADHD and other forms of discrimination toward the child and family; (3) empowering caregivers to advocate for their child and seek treatment; (4) addressing unmet needs and learning to navigate systems of care, (5) building trusting relationships with treatment providers, and (6) engaging children in care and empowering them for a future with ADHD.

Equity-Centered Intervention Content Development

All materials were newly developed based on our previous qualitative work (Spencer et al., 2021), centering the voices and perspectives of caregivers and children who experience inequities in ADHD care in the content. We created content in five modules outlined in Supplemental 2 to be completed in approximately five to seven weekly sessions, with each

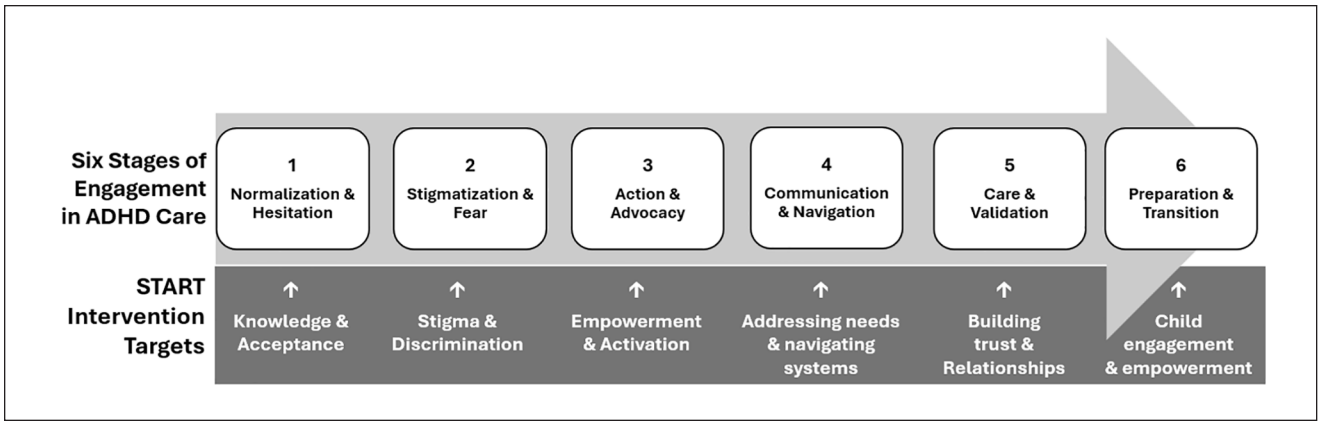


Figure 1. Selected intervention targets to facilitate the Six Stages of Engagement in ADHD Care, adapted from Spencer et al., 2021.

session planned for 1 hr, via either in-person or virtual sessions. Modules 1 to 4 were for the caregiver(s) alone, while module 5 was for caregiver(s) and child together. We developed an interventionist manual for all modules including scripted language for each section to improve fidelity of delivery, as well as guidance on personalizing the content based on a family's goals, preferences, and stage of engagement. Each module was designed such that it could be personalized to address any intervention target and thus facilitate any stage of engagement, and questions were embedded in the manual to assist the interventionist with this personalization process. Supplemental 3 outlines how content related to each intervention target was included in all modules. We also developed a companion caregiver and child guide with over 40 new handouts including texts and graphics. For example, we created the "4-Gear Model" to describe a multifaceted approach to treating ADHD with medications, school accommodations, behavioral therapy, and caregiver wellness. Handouts incorporated perspectives from caregiver and youth interviews, including de-identified quotes and paraphrased advice to illustrate how other caregivers and youth described ADHD symptoms, how they responded to ADHD stigma, and how caregivers talked to their children about ADHD. We distilled caregiver advice about facilitating caregiver-child discussions about ADHD into a framework we called PLANS (Prepare and Practice; Listen and Ask; Explain ADHD with Analogies; Normalize the Negatives; Support and Succeed), introduced in module 4 to guide the discussion with the child in module 5.

Community Health Worker Training

Community Health Workers (CHWs) were bachelor's or master's level individuals who were not health care professionals, but had cultural congruence with our patient population, personal experience with ADHD or mental health conditions, or personal experience working with children and families in the community. The principal investigator (AS), a child psychiatrist with expertise in ADHD, conducted 12 hr of interventionist training, which included: (1) ADHD basics and beyond; (2) review of the Six Stages framework and intervention development process; (3) detailed review of the intervention manual and handouts; (4) motivational interviewing skills; (5) privacy, confidentiality, and safety; and (6) role plays. Prior to beginning intervention delivery, we assessed initial competence via role plays and ongoing competence via weekly supervision with the Principal Investigator throughout intervention delivery. Intervention delivery, training, and supervision were predominantly conducted virtually due to the COVID-19 pandemic.

Intervention Feasibility Testing

We conducted a single-arm adaptive pilot test of START to evaluate feasibility and acceptability and engagement of intervention targets, and to obtain feedback to refine the intervention in preparation for a randomized controlled trial. Feasibility testing was approved by the Boston University Medical Center Institutional Review Board.

Participant Recruitment

We recruited participants from pediatric treatment settings at an urban, academic safety net hospital via direct clinician referrals and behavioral health waiting lists. Families were eligible if they were English-speaking and had a child ages 6 to 12 years old either with newly diagnosed ADHD in the past 3 months, suspected ADHD formally diagnosed by the PI, or previously diagnosed ADHD with no treatment since diagnosis or in the past 12 months. Children with psychiatric comorbidities were included except psychosis and intellectual disability because these conditions would substantially change the treatment approach.

Quantitative Measures

Quantitative measures were administered at baseline and then following the last intervention session or 4 months after enrollment, whichever came first. Feasibility was assessed with process measures including enrollment and retention rates, as well as average session length and time to intervention completion. Caregiver-reported feasibility and acceptability was assessed at intervention completion with a 17-item *Feasibility and Acceptability Questionnaire* (Supplemental 4a) developed by the study team using a 5-point Likert Scale from Strongly Agree to Strongly Disagree. We assessed fidelity through interventionist tracking of session time, session duration, and module completion, as well as monitoring of interventionist session reports at weekly supervision meetings. Finally, we assessed engagement of intervention targets with our post-intervention Intervention Objectives Questionnaire to understand whether the content was delivered as intended. We assessed whether we addressed the selected intervention targets with three questionnaires also developed by the study team (Changes attributed to START, the Intervention Objectives Questionnaire, and the ADHD Knowledge Questionnaire). Administered only at intervention completion, *Changes Attributed to START* (Supplemental 4b) surveyed caregivers about changes in treatment engagement with nine items along a 4-point Likert scale. The final three questionnaires were administered both at baseline and intervention completion. The *Intervention Objectives Questionnaire* (Supplemental 4c) included 17 items assessing caregiver-report about treatment engagement based on the five pre-determined intervention targets (knowledge, stigma, activation, needs, trust, child engagement) along a 5-point Likert Scale from 1 = Strongly Disagree to 5 = Strongly Agree. The *ADHD Knowledge Questionnaire* (Supplemental 4d) included 10 items assessing ADHD knowledge using multiple choice or true/false format. Finally, the impact of START on treatment utilization was assessed using a *Treatment Utilization Questionnaire* (Supplemental 4e) with questions adapted from the National Survey of the Diagnosis and Treatment of ADHD and Tourette Syndrome (NS-DATA). (*SLAITS - National Survey of the Diagnosis and Treatment of ADHD and Tourette Syndrome*, 2022) These questions asked about both lifetime and current receipt of medication for ADHD, outpatient therapy (including family, individual, and behavioral therapy), school services (including IEP or 504), partial hospitalization, and inpatient psychiatric hospitalization.

Qualitative Measures

Exit interviews were conducted with participants to elicit feedback on participants' experiences with the intervention. All participants were invited to conduct exit interviews regardless of whether they completed the intervention. Research staff who were not involved in intervention delivery conducted interviews using a structured interview script, which is available as Supplemental 5. We refined START based on feedback from these post-intervention surveys and exit interviews.

Data Analysis

To evaluate feasibility, we calculated the percent of enrolled families who completed research questionnaires and the intervention separately. We calculated the median intervention duration in weeks, median number of sessions, and median average time per session. We also calculated these process measures separately for participants who completed versus those who did not complete START. On the post-intervention questionnaires *Feasibility and Acceptability* and *Change Attributed to START*, we calculated the percent of participants who indicated "agree" or "strongly agree" with each item along the Likert scale. To compare pre-intervention and post-intervention questionnaires, we used the sample with complete pre-post data ($N = 16$). On the ADHD Knowledge Questionnaire, we calculated the median percent correct with interquartile range (IQR) before versus after the intervention. On the Intervention Objective Questionnaire, we calculated the percent of participants reporting "agree" or "strongly agree" for each item as well as the median score along the 5-point Likert scale from strongly disagree to strongly agree before versus after the intervention. For treatment utilization, we calculated the percent of children currently receiving each type of outpatient treatment (any medication for ADHD, any therapy, and any school supports) as well as how many of these outpatient treatment categories (from 0 to 3) a family received before versus after the intervention. We conducted non-parametric Wilcoxon signed-rank tests to compare the median difference between scores on pre-post surveys before and after START, setting alpha at $p < .05$. We conducted the non-parametric McNemar's test to compare the percent of children receiving each type of outpatient treatment before and after START. For qualitative analyses, in-depth interviews were recorded, transcribed, checked, de-identified, and analyzed via thematic analysis. The study team met as a group to review five initial interviews and develop an initial codebook, which was applied independently by two separate coders to all interviews. The study team then met to review themes and identify relationships among themes. Coding and thematic analysis was facilitated by Dedoose software (*Dedoose. 9.0.107. Web Application for Managing, Analyzing, and Presenting Qualitative and Mixed Method Research Data*, 2023).

Results

Intervention Feasibility Testing

Sample Demographics. We screened 43 families, of which 37 were eligible, 18 consented to participate and began the

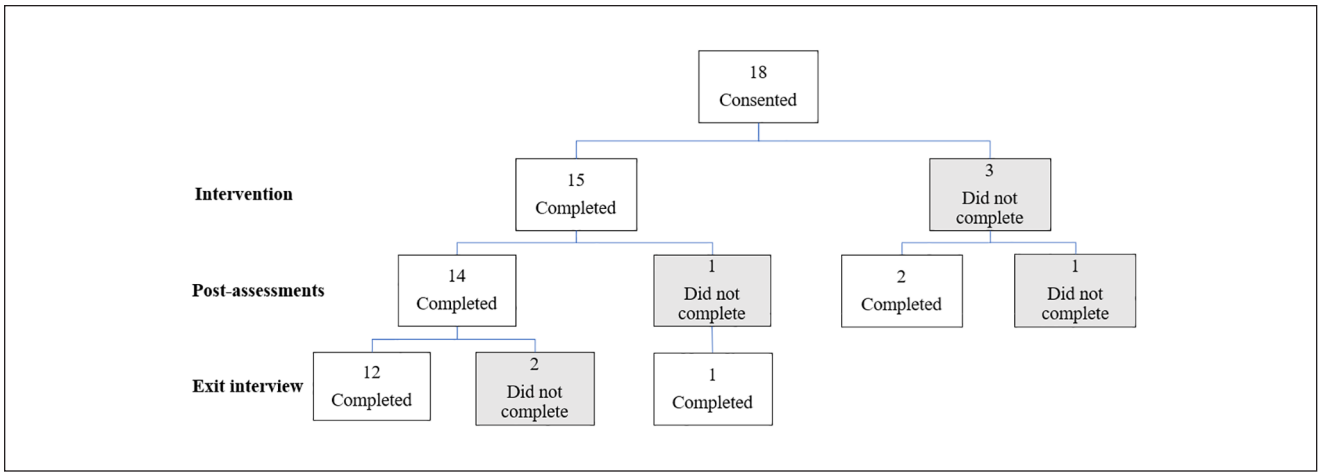


Figure 2. Flow chart of participants in the feasibility test of the START ADHD Engagement Intervention.

study, 15 (83%) completed START, 16 completed all study questionnaires, and 13 completed the exit interview. Figure 2 shows a flowchart of participant retention following enrollment. Participants who completed post-intervention questionnaires were included in analyses ($n = 16$) regardless of intervention completion.

Demographics are shown in Tables 1 (children) and 2 (caregivers). Caregivers had a mean age of 36 ± 6 years, almost all (94%) identified as female, 39% were married, and 22% were not married but living with a partner. Most caregivers identified as Black/African American/Haitian (56%) or Hispanic (17%) and 34% spoke a language other than English at home. Just over half of caregivers (56%) had attained some postsecondary education, 61% were employed, and 39% reported difficulty covering basic needs. Children had a mean age of 7.4 ± 1.3 years, 72% were male, 50% identified as Black or African American, and 83% were covered by public insurance.

For all 18 participants who began the intervention, the median number of sessions completed was 6 (IQR = 5–6) and median intervention duration was 5 weeks (IQR = 4–8). The 15 participants who completed all modules of START did so in a median of 6 sessions (IQR = 5–7) in 6 weeks (IQR = 5–8), whereas the three others completed a median of three sessions (IQR = 3–4) in 3 weeks (IQR = 2–4). The median average time per session across participants was 60 min (IQR = 54–66). For the 16 participants who completed pre- and post-intervention questionnaires, the median time between pre- and post-intervention questionnaire completion was 81 days (IQR = 50–98).

Figure 3 shows detailed caregiver-reported feasibility and acceptability results. All caregivers reported that they were satisfied with the quality of the intervention and materials provided, and that they would recommend START to others. Almost all caregivers (88%) agreed or strongly agreed that the intervention was helpful. However, half of caregivers (50%) reported that START was too short, while only one caregiver (6%) reported that START took too much time. A few caregivers reported that there was too much information (25%). These results identified some areas for improvement.

On the 9-item Changes Attributed to START questionnaire administered post-intervention, 94% of caregivers reported that START made them feel more confident in seeking treatment for ADHD, and many caregivers reported that START made them more likely to consider medications (69%), therapy (75%), and school services (88%) for their child's ADHD. Most caregivers reported that START changed how they will talk with others about ADHD including their child (88%), family and friends (88%), and their child's doctor (81%). However, only 44% of caregivers reported that START helped them to overcome financial barriers to care.

There was a statistically significant increase in total outpatient treatment utilization scores from a median of 1.0 (IQR = 1.5) pre-intervention to 2.0 (IQR = 0.5) post-intervention ($p = .0034$). The percent of children receiving each type of outpatient treatment service increased from pre- to post-intervention, including an increase in medication use (from 38% to 50%, $p = .1573$), therapy use (from 31% to 69%, $p = .0578$), and a statistically significant increase in receipt of school accommodations (from 38% to 88%, $p = .0047$). The increase in therapy was mostly due to families newly receiving behavioral therapy, which increased from 0% pre-intervention to 31% post-intervention. On the Intervention Objectives Questionnaire, more parents agreed or strongly agreed with each objective on the post-intervention questionnaire compared to pre-intervention, and there was a statistically significant ($p < .05$) increase in scores on the 5-point Likert scale for 16 of 17 intervention objectives (Figure 4). There was a statistically significant increase in ADHD Knowledge scores from a median of 65% correct

Table 1. Characteristics of Caregiver Participants in the Pilot Feasibility Test of START.

Characteristics of Caregiver Participants	*N* = 18
Parent age (years): mean ± *SD*	36 ± 6
Sex: *n* (%)	
Female	17 (94)
Male	1 (6)
Parent relationship to child	
Biological parent	17 (95)
Marital status: *n* (%)	
Married	7 (39)
Not married, but living with a partner	4 (22)
Never married	6 (33)
Don't know/refused	1 (6)
Birthplace: *n* (%)	
United States	11 (61)
Dominican Republic	1 (6)
El Salvador	1 (6)
Guinea	1 (6)
Haiti	3 (17)
Jamaica	1 (6)
Race: *n* (%)	
Black/African American	8 (44)
Haitian	2 (11)
Hispanic, Latinx, or Spanish Origin	3 (17)
White/Caucasian	3 (17)
American Indian/Alaskan Native	1 (6)
More than one race	1 (6)
Education level: *n* (%)	
Less than high school degree	4 (22)
High school degree/GED	4 (22)
Postsecondary education	5 (28)
College degree or higher	5 (28)
Primary language spoken at home: *n* (%)	
English	12 (67)
Spanish	2 (11)
Haitian Creole	3 (17)
Fulani	1 (6)
Employment status: *n* (%).	
Employed	11 (61)
Unemployed	7 (38.9)
Social risks	
Difficulty covering basics like food and housing	7 (39)
Childcare affecting work or school	11 (61)
Difficulty paying mortgage or rent	8 (44)
Utility Risks (gas/electricity/oil)	6 (33)
Food insecurity	4 (22)

(IQR = 30) pre-intervention to 90% (IQR = 15) post-intervention ($p < .0001$).

Qualitative Results

Thirteen caregivers completed exit interviews. Thematic analysis revealed both positive and constructive feedback related to each intervention target in the Six Stages framework.

Caregivers described improved understanding and acceptance of an ADHD diagnosis (Stage 1) after START.

> *It helped me understand a lot more about ADHD and the effects of it and realizing that it's not one thing, but maybe contributing things that can be involved with a child having ADHD and it's not bad parenting. . . it's not about diet. . . there's a lot of things that go into it, the biology of it, just a lot of things that I didn't understand.*

Some caregivers commented on the personalized content and appreciated information communicated in the words of other parents with similar experiences.

> *I learned a lot. I've learned things like how I'm not alone. I liked how it was personalized. For example, like things that. . . parents who have children with ADHD or how kids who have ADHD feel. Like, it's not in technical terms, but is more like, in their words like how the everyday person would say it.*

However, a few caregivers reported that they knew much of the information in early modules and recommended a more personalized approach.

> *Certain things. It was just like, I already know this, I already know this. That part was just like, not helpful. . .after going for third visit fourth, fifth, we started getting into things like, oh, I really didn't know this.*

Caregivers reported that the intervention helped them confront their own stigma about ADHD, as well as stigma from others (Stage 2). We received overwhelmingly positive feedback about the content and strategy addressing stigma.

> *I learned a lot of ways to explain ADHD. . .to his other family members who kind of frown upon this diagnosis in general. Because a lot of times I will try to explain to them, you know why I'm making some of the choices I'm making, and they're just trying to poke holes and all that. So it really helped me be able to counteract those kinds of conversations.*

After participating in START, caregivers described an improved understanding of treatment options for ADHD and willingness to consider these options (Stage 3), although in multiple cases there was ongoing hesitancy about medication.

> *I probably wasn't. . .big in support of medications. So, I think that it's helped me learn a lot about that. And the fact is that I want my son to be the best he can be. And if it requires a medication, I think it's helped me understand more that it's not like, he's going to get addicted, or it's going to do something to his personality. And that we can try things to make sure that it's the right fit for him. . .he hasn't been on put on meds yet. And we're going to wait a little bit, but I think that I'm more open to*

Table 2. Demographic and Clinical Characteristics of Child Participants in the Pilot Feasibility Test of START.

Characteristics of Child Participants	*N* = 18
Gender (%)	
Male	13 (72)
Female	5 (28)
Age (years): Mean ± *SD*	7.4 ± 1.3
Race: *n* (%)	
Black/African American	9 (50)
Hispanic, Latinx, or Spanish Origin	3 (17)
White/Caucasian	2 (11)
More than one race	4 (22)
Child birthplace: *n* (%)	
United States	15 (83)
Belgium	1 (6)
Jamaica	1 (6)
Other psychiatric and developmental conditions: *n* (%)	
Developmental delay	4 (24)
Behavioral or conduct problem	2 (11)
Anxiety	2 (11)
Speech or other language	3 (17)
Autism or other autism spectrum disorder	1 (6)
Medical conditions: *n* (%)	
Asthma	3 (17)
Hearing	1 (6)
Vision	1 (6)
Food allergies	1 (6)
Health insurance: *n* (%)	
Public or health safety-net	15 (83)
Commercial/private	3 (17)

understanding that and maybe if we have to, it might be something and that helped me -- the intervention helped me understand that.

Caregivers emphasized the benefit of learning about the system of care for ADHD and help with navigation to services (Stage 4).

[START helped with] being able to find the behavioral specialists that help the most, because number one like I said, I wouldn't even know where to look. . .obviously I have a certain criteria like I want this person to be well versed in ADHD in the first place. So that makes it even harder to find someone who specializes in that. So having the help in the program, like [the CHW] reaching out and providing me of resources was probably the best part.

However, caregivers provided feedback that support beyond the end of START would have been helpful for ongoing care navigation and information.

What kind of game plan should I have in place. . . you know, to keep my questions answered and making sure I'm getting the most up-to-date information, valid information from somebody, you know?

Caregivers described feeling validated and supported in their decision to seek and stick with care for their child's ADHD due to START (Stage 5).

Some of it is just needing to hear that you're doing the right thing because you just don't know. There's no guidebook, there

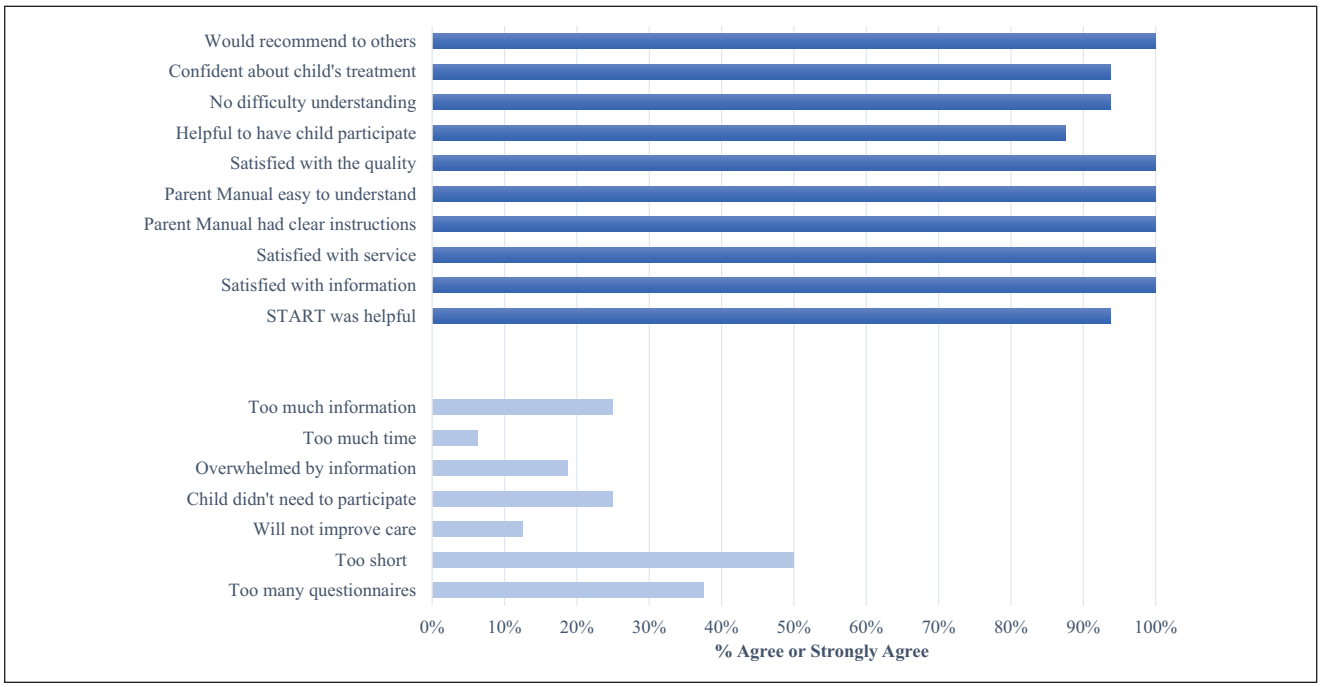


Figure 3. Percent of participants responding "agree" or "strongly agree" to each item on the Feasibility and Acceptability Questionnaire. The graph first presents the "positively worded" items followed by the "negatively worded" items.

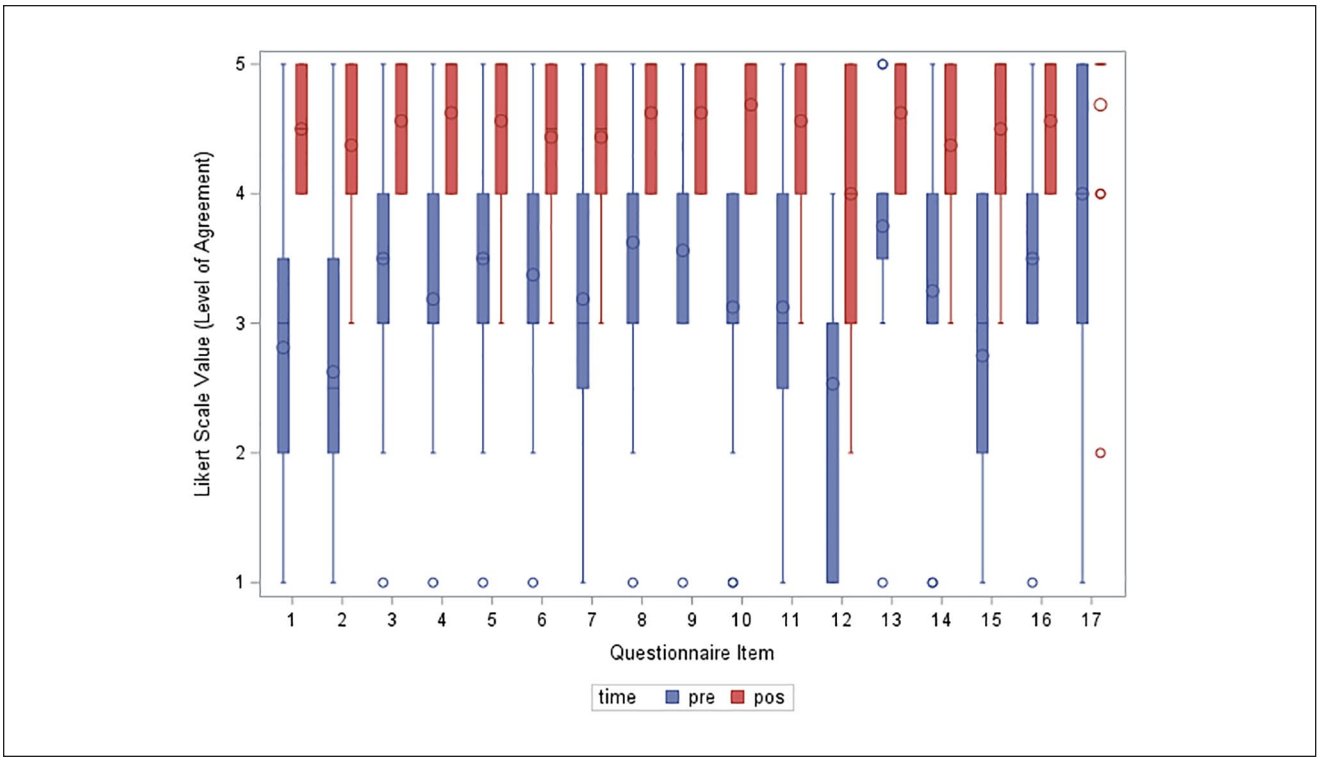


Figure 4. Median Likert scale score for each item on the Intervention Objectives Questionnaire (measured from 1 = strongly disagree to 5 = strongly agree) before and after the START ADHD Community Health Worker intervention.

> *are no instructions when you get that kind of diagnosis that guides you and tells you what you need to do. So you're doing your best but you don't know if you're doing the right thing. . . or if there's more you could be doing. So this really helped me narrow in and focus on areas that I needed help with. . .But it also helped me, you know, validate that I was doing the right thing too.*

Caregivers noted that talking to their child about ADHD was often challenging, but that START provided helpful support around approaching this conversation (Stage 6).

> *There was a list of different people who have ADD. . . they got the sports players but like he knew some of the authors, he knew some of the, you know, actors, stuff like that. So, he was like enlightened by that. . . . It doesn't mean that he thinks he's broken. Well, you can do great things and that was eye opening to him. . . He's like amazed of the brain that he had. So, like we just kept pointing different people out and that was huge for him.*

However, some parents reported that because of their child's age, developmental level, or attention span, it was difficult to engage their child with the content in module 5.

> *For me, I feel like he's – he's six years old. It's – it's kind of too early for him to make it – even though he still tried different way. . .I told her that he like cars. We try with cars, with cartoon character, he just don't wanna hear about it.*

Intervention Refinement

Based on these results, we made refinements to START, briefly summarized here. We expanded the personalization strategies by reviewing scores on baseline questionnaires with the CHW prior to initiation of the intervention and adding questionnaires to more formally assess goals and stigma (e.g., the ADHD Preference and Goal Instrument; Fiks et al., 2012). In addition, the manual and training materials now instruct CHWs to review the full table of contents with caregivers during the initial goal-setting session, to identify areas of greatest interest. CHWs are now also instructed to review family preference on session length, number, and timing (e.g., multiple per week vs. every other week) ahead of time and to accommodate families within reason. We added more content about medication stigma, comorbid psychiatric symptoms and disorders, and learning disorders in modules 2 and 3, to both the CHW manual and the parent booklet. We expanded sections with frequently asked questions that were previously in the CHW manual only, and moved them into the parent booklet, to increase the amount of written information provided directly to families. We added a list of local resources to facilitate

connections in module 4. We added more guidance for CHWs before module 5 ("Talking to your child and preparing for the future") to assist them with adjusting the content for the child's developmental stage, increased the guidance provided on this module in supervision sessions, and increased the time planning with parents for this session. We also expanded our list of famous people with ADHD to apply across the developmental spectrum, decreased the expectation for amount of time spent with the child, and included an expectation that this module is completed in person. We created a new module 6 to ensure that needed referrals had been made and support planning for ongoing treatment.

Discussion

We report on the development of a novel Community Health Worker (CHW) intervention for ADHD—SupporT for ADHD and Related Treatment (START)—to promote engagement in care. START is based on perspectives of racial and ethnic minoritized families documented in prior qualitative work with input from a Community Advisory Board. Here, we tested START for feasibility and engagement of intervention targets in a single arm pilot. Retention in the intervention was high; caregivers reported high acceptability; knowledge about ADHD and other intervention objectives improved; and treatment utilization for ADHD increased from before to after START. These preliminary findings support the possibility that START could improve engagement and increase utilization in care for racial and ethnic minoritized families who face many barriers to care for ADHD.

CHW interventions have great promise to reduce inequities in receipt of care for ADHD as they have been used successfully for other pediatric medical conditions (Raphael et al., 2013), but this treatment model is underdeveloped and understudied for ADHD. There have been a few studies incorporating bachelor's level care managers(Fiks et al., 2017; Guevara et al., 2021; Silverstein et al., 2015) to provide care coordination and, in some cases, brief engagement interventions for ADHD in primary care. For example, Guevara et al. conducted a randomized controlled trial of care management combined with a patient portal versus portal alone for children with ADHD (Guevara et al., 2021). Care management involved quarterly communication with parents, teachers, and clinicians, care coordination, and psychoeducation, while the portal facilitated goal-setting, communication, and information transfer. This study found reductions in ADHD symptoms in both arms with no statistically significant difference, and authors noted that inconsistent engagement with the care manager limited the potential additive benefit over the portal. Silverstein et al. (2015) conducted a randomized controlled trial of two care management strategies for suspected ADHD in the pediatric collaborative care setting. Participants with suspected ADHD were assigned care managers, and an enhanced care arm incorporated additional training for care managers in motivational interviewing and parent management techniques (Triple P). The study showed improved symptom trajectories in the enhanced care group only for children ultimately diagnosed with ADHD (40%) and supports a precedent for the role of a bachelor's level team member in treatment of ADHD in primary care. Waxmonsky et al. (2023) report results of single-arm study testing a two-stage engagement intervention for ADHD delivered by care coordinators in primary care that included an assessment battery, manualized content using shared decision-making and motivational interviewing principles, and referrals to care. Intervention completers in the study had higher rates of receiving school-based supports and counseling, although the study was limited by low rates of engagement in the intervention.

To our knowledge, only one study has reported on paraprofessional delivery of behavioral parent training for ADHD. Chacko et al. describe an open pilot trial of paraprofessional-delivered behavioral parent training for ADHD with seven parents (Chacko et al., 2020). Paraprofessionals were Family Peer Advocates from an affiliate organization to a developmental pediatric practice, supervised by a pediatric psychologist. They report high parent engagement and satisfaction with the intervention, which was found to be feasible and delivered with high fidelity, and it led to improvements in ADHD symptoms and child functional impairment. Preliminarily, this study also provides support for the idea that community-based paraprofessionals may be a feasible and effective solution to address disparities in ADHD treatment by participating in the delivery of evidence-based care.

The published literature includes a few other psychoeducational and engagement-focused interventions for ADHD that show some promise for improving ADHD symptoms or increasing treatment utilization when delivered by a mental health professional. Ferrin et al. reported results of a randomized controlled trial of a six-session psychoeducation group led by doctoral level clinicians (psychiatrists and psychologists) in the UK ($N=69$). The intervention led to improvements in ADHD symptoms 6 months later compared to a control group (Ferrin et al., 2020). Another ADHD engagement intervention called Bootcamp for ADHD (BC-ADHD) is a four session, clinician-delivered, group-based intervention for parents to increase use of evidence-based ADHD treatment (Nissley-Tsiopinis et al., 2023; Normand et al., 2024). In an open trial ($N=59$), BC-ADHD was feasible to deliver and acceptable to families, with increases in caregiver empowerment, reduced concerns about behavioral and medication treatment, and increased receipt of treatment post-intervention. While

these interventions have promise, adaptation may be needed for socioeconomically disadvantaged families and racial and ethnic minoritized families to improve both feasibility and acceptability, and CHW delivery could improve accessibility. There is other evidence for successful development of family-centered interventions for increasing caregiver engagement in their child's mental health care. For example, Thomas et al. included parent input in the development of a group psychoeducation intervention for Latino parents of children with mental health needs (Thomas et al., 2017). The intervention, MePrEPA, facilitated by bilingual doctoral-level clinical psychology graduate students, improved parent activation and school interactions in navigating their children's mental health needs. Our pilot suggests that our CHW intervention may also improve parent activation and self-efficacy, as well as knowledge, intention to seek care, and receipt of treatment, but further testing is needed.

To our knowledge, there are no existing evidence-based interventions explicitly addressing stigma in ADHD care although it is an important barrier to care (Schoeman & Voges, 2022). Because of their lived experience in communities, CHWs are well-positioned to build trust and address stigma as a barrier to care. Our strategy to support stigma reduction in START materials includes communicating accurate information, helping families recognize ADHD stigma, and supporting families to address stigmatizing statements and attitudes as they seek care for their children. These strategies and corresponding materials were informed by racial and ethnic minoritized families interviewed as part of our previous qualitative work during intervention development. Families reported increases after start in their understanding about ADHD, ability to recognize ADHD stigma, and comfort communicating with others about ADHD. Our study has some limitations. To test intervention feasibility, we conducted a single arm pilot study, not powered for hypothesis testing nor designed for effectiveness testing with clinical outcomes. Thus, we cannot make conclusions about whether START will lead to increased engagement, treatment utilization, and clinical improvement compared to a control group, which must be tested in further research. However, we did determine that our theory-based, family-based, and equity-centered intervention was feasible with high retention and acceptability to families and had excellent evidence of engaging intervention targets, and we refined the intervention based on family feedback. We developed and pilot tested intervention materials only in English, although formative research informing the intervention approach was based on interviews conducted in other languages as well, and we have already completed an adaptation to Spanish for the next study. Considering that the goals of the intervention are to improve equitable treatment of ADHD, translating and adapting our intervention into other languages will be critical. Although we did have some strategies to assess fidelity, we did not record intervention sessions to independently assess fidelity to specific module content. We are using this more intensive strategy to assess fidelity in the subsequent randomized study. Furthermore, competence of CHWs was more subjectively assessed here via role-plays and ongoing supervision, but we have added a pre- and post-training knowledge assessment for CHWs in our randomized study. We also noted that in this study, the time between completion of pre- and post-intervention sessions was highly variable, and thus we have modified the research protocol in the subsequent randomized trial to send post-intervention questionnaires at 3 months after baseline questionnaires to all participants. We did not conduct a cost-benefit analysis of the intervention; this would be an important step for a future study to understand whether a six-session community health worker intervention is a cost-effective strategy for improving engagement in ADHD care. Finally, families identified some areas of growth for START, including improved personalization based on baseline knowledge of ADHD and the child's developmental level.

Despite these limitations, we present promising results from the development and pilot testing of an equity- and family-centered Community Health Worker intervention for ADHD—START—to reduce disparities in care. We describe a method for Community Health Worker intervention development that centers the voices of racial and ethnic minoritized families and their children with ADHD that could be translated to other mental health conditions. Results from our pilot test suggest that START is feasible, acceptable to families, and has promise for improving interest and engagement in ADHD care. Future research should evaluate the effectiveness of START in a randomized clinical trial to promote engagement in care and improve clinical outcomes for racial and ethnic minoritized children with ADHD.

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Supplemental Material

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